iii) Pythagorean Trignometric identities:

$$\sin^2 A + \cos^2 A = 1$$

$$1 + \tan^2 A = \sec^2 A$$

$$1 + \cot^2 A = \text{cosec}^2 A$$

Now, let us look at the differentiation rules for trigonometric functions.

$$D(\sin A) = \cos A$$

$$D(\cos A) = -\sin A$$

$$D(\tan A) = \sec^2 A$$

$$D(\cot A) = \text{cosec}^2 A$$

$$D(\sec A) = \sec A \tan A$$

$$D(\text{cosec}) = -\text{cosec}\, A \cot A$$

Some generalised formulae for integration:

Integral Rule	**General Rule**
$\int \cos x dx = \sin x + C$	$\int \cos(ax+b) dx = \frac{1}{a}\sin(ax+b) + C$
$\int \sin x dx = -\cos x + C$	$\int \sin(ax+b) dx = -\frac{1}{a}\cos(ax+b) + C$
$\int \tan x dx = -\ln\|\cos x\| + C$	$\int \tan(ax+b) dx = -\frac{1}{a}\ln\|\cos(ax+b)\| + C$
$\int \cot x dx = \ln\|\sin x\| + C$	$\int \cot(ax+b) dx = \frac{1}{a}\ln\|\sin(ax+b)\| + C$
$\int \sec x dx = \ln\|\sec x \tan x\| + C$	$\int \sec(ax+b) dx = \frac{1}{a}\ln\|\sec(ax+b) + \tan(ax+b)\| + C$
$\int \text{cosec}\, x dx = \ln\|\text{cosec}\, x + \cot x\| + C$	$\int \text{cosec}(ax+b) dx = \frac{1}{a}\ln\|\text{cosec}(ax+b) + \cot(ax+b)\| + C$

Example 12.14 : Evaluate the integral $\int \cos(15x)\cos(4x) dx$

Ans: You may notice that we have utilised trigonometric identities to convert products to sums

$$\int \cos(15x)\cos(4x) dx = \frac{1}{2}\int \cos(11x) + \cos(19x) dx =$$

$$\frac{1}{2}\left(\frac{1}{11}\sin(11x) + \frac{1}{19}\sin(19x)\right) + C$$

Example 12.15: Evaluate $\int \cos x \sin^5 x dx$

Ans: Put u = sin x so that du = cos x dx

Thus, $\int \cos x \sin^5 x dx = \frac{1}{6}u^5 du = \frac{1}{6}\sin^6 x + C$

Example 12.16: Evaluate the following integral: $\int \sin^5 x dx$

Ans: $\int \sin^5 x dx = \int \sin^4 x \sin x dx = \int (\sin^2 x)^2 \sin x dx = \int (1-\cos^2 x)^2 \sin x dx$

Put u = cos x to get

$\int \sin^5 x dx = -\int (1-u^2) du = \int (1-2u^2+u^4) du = -\left(u - \frac{2}{3}u^3 + \frac{1}{5}u^5\right) + C$

$= \left(-\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + C\right).$

Example 12.17: Evaluate $\int (3\sin x - 4\sec 2x) dx$

Ans: $\int (3\sin x - 4\sec 2x) dx\, 3\sin x dx - \int 4\sec 2x,$

Or, $3\int \sin x dx - 4\int \sec 2x dx = 3\int \sin x dx - 4\int \sec^2 x dx = -3\cos x - 4\tan x + C$.

Example 12.18: Evaluate $\int \cos(2x-6) dx$

Ans: $\int \cos(2x-6) dx$

Put $u = 2x - 6$ so that $\frac{du}{dx} = 2$ or, $dx = \frac{1}{2}du$

We now have $\cos(2x-6)dx = \int \cos\frac{1}{2} du = \int u^2 du = \frac{-u^3}{3} + C = -\frac{\cos^3 x}{3} + C$.

12.5 INTEGRATION BY PARTS

A method of integration is often useful when two functions are multiplied together. Let us return to differentiation of product of two functions $u = f(x)$ and $v = f(x)$ which gives,

$d(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

From this, we can obtain

$\int (uv)' dx = \int uv' dx + \int vu' dx$

Because $\int (uv)' dx = uv$

$\therefore uv' dx = uv - \int vu' dx$

Example 12.19: Solve $F(x) = \int xe^{-x} dx$

Ans: Let $u(x) = x, v(x) = -e^{-x}$

So that $u'(x) = 1, v'(x) = e^{-x}$

$\therefore f(x) = \int u.v' dx$

$$= uv - \int vu'dx$$

$$= -xe^{-x} + \int e^{-x}\,dx$$

$$= -xe^{-x} - e^{-x} + c = -e^{-x}(1+x) + c$$

Example 12.20: Evaluate the following integral: $\int (3t+5)\cos\left(\frac{t}{4}\right)dt$

Ans: Put $u = 3t+5$ and $dv = \cos\left(\frac{t}{4}\right)dt$

So that du = 3t and $v = 4\sin\left(\frac{t}{4}\right)$

The integral is then,

$$\int (3t+5)\cos\left(\frac{t}{4}\right)dt = 4(3t+5)\sin\left(\frac{t}{4}\right) - 12\int \sin\left(\frac{t}{4}\right)dt = 4(3t+5)\sin\left(\frac{t}{4}\right)$$
$$+ 48\int \cos\left(\frac{t}{4}\right) + C.$$

Example 12.21: Integrate $\int x\ln x\,dx$

Ans**:** Let $u = \ln x$ and $dv = x\,dx$ so that $du = \frac{1}{x}dx$ and $v = \frac{x^2}{2}$.

Therefore, $\int x\ln x dx = \frac{x^2}{2}\ln x - \int \frac{x^2}{2}\frac{1}{x}dx = \frac{x^2}{2}\ln x - \frac{1}{2}\int x dx =$

$$\frac{x^2}{2}\ln x - \frac{1}{2}\frac{x^2}{2} + C = \frac{x^2}{2}\ln x - \frac{1}{4}x^2 + C.$$

Check Your Progress 2

1) Since indefinite integration is the anti-derivative, how do you show that $\int \cos ax dx = \frac{1}{a}\sin ax + C$.

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2) While solving the integration problem, how do you decide to use a) substitution and b) integration by parts?

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3) Use the formula for integration by parts to evaluate $\int x^4 \ln x dx$. Why $\ln x$ would be the best choice for u?

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12.6 SOME USEFUL FORMULAE

Here are some useful formulae for integration.

1) $\int x^n dx = \frac{x^{n+1}}{n+1} + c, n \neq -1$

2) $\int x^{-1} dx = \int \frac{dx}{x} = \log x + c$ for any $x > 0$

3) $\int e^{mx} dx = \frac{e^{mx}}{m} + C$ for any m

4) $\int a^{mx} dx = \frac{a^{mx}}{m \log a} + C$

5) $\int \cos ax dx = \frac{\sin ax}{a} + c$

6) $\int \sin ax dx = -\frac{\cos ax}{a} + c$

7) $\int [K_1 f(x) + K_2 g(x) dx] = K_1 \int f(x) dx + K_2 g(x) dx + c$ for $K_1, K_2 \geq 0$

12.7 DETERMINATION OF THE CONSTANT OF INTEGRATION

We have noted that the indefinite integral always contains an arbitrary constant. The value of this constant can be precisely determined (and the integral rendered unique thereby) if it is known that the integral function $y = f(x) + c$ obeys some prescribed **initial condition** $(y = y_0$ when $x = 0)$ or more generally **boundary condition** $(y = y_0$ when $x = x_0)$. To illustrate, we know $\int (x^2 + 3) dx = \frac{1}{3} x^3 + 3x + c$

Here $y = f(x) + c = \frac{1}{3} x^3 + 3x + c$. Suppose, we are given the initial condition, y = 20 when x = 0. Then

$$y_0 = f(0) + c = c = 20$$

This fixes the integral as the unique function $y = \frac{1}{3} x^3 + 3x + 20$.

12.8 DEFINITE INTEGRALS

When we use integration to arrive at a number, definite integral helps. It has both the start and the end values: in other words, we operate in an interval. This type of integration is useful in finding areas, volumes and central points.

12.8.1 Finding Area

We look at the *Area Problem* before we deal with definite integrals. It will help us interpret definite integral and lead us to the definition of the definite integral. Suppose we have a function f(x) that is positive on some interval [a, b]. If we want to determine the area of the region between the function and the x-axis, we do this in two ways: using an approximation (finding areas of rectangles) and using integration, where the latter is also a method of approximation, of course, by dividing the area under consideration into very large number of very small rectangles/ areas.

For the method of approximation, the space under the function is divided into rectangles and adding areas of those rectangles we get the area under the curve (Fig. 12.2). For example, to determine the area under the curve, that is, the shaded region below we divide the area into rectangles.

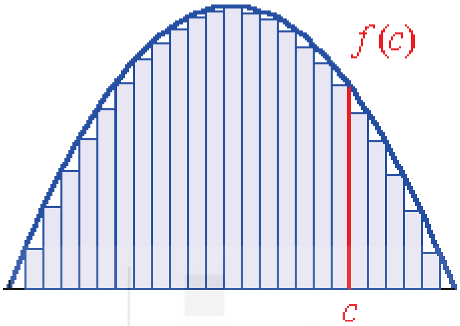

Fig. 12.2: Finding area using inner rectangles

The height of each rectangle is found by calculating the function values, as shown for the typical case x = c. We get a better result if we take more and more rectangles. In Fig. 12.2, we are approximating the area using **inner rectangles** (each rectangle is inside the curve).

We could also find the area using the **outer rectangles.**

To understand the process let us estimate the area by dividing up an interval, [a, b] into n subintervals each of width, $\Delta x = \frac{b-a}{n}$. Then construct a rectangle whose height is given by the function value at a specific point in the interval. The area of each of these rectangles can be derived by $\Delta x . h_0$ where h is the height of the rectangle in that interval. When we add each of these, we get an estimate of the area under the curve using three options of right, left and the middle value of the rectangles of the curve.

Suppose you want to determine the area of the function $f(x) = x^2 + 1$ in the interval [a, b] by considering 4 subintervals. Fig. 12.3 shows the area under examination.

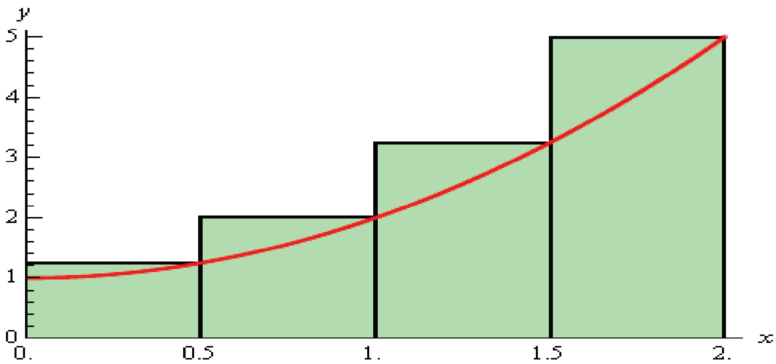

Fig. 12.3: Finding area using outer rectangles (right end points)

From Fig. 12.3 it is apparent that by choosing the height of each of rectangle overestimates the area. The reason is that each rectangle takes in more area stretching beyond the value of the function as shown by the curve in each case.

Since the width of each of the rectangles is ½ and the height of each rectangle at the value given by the function at the right endpoint of the concerned interval $f(x_i)$ the estimated total area is

$$A_r = \frac{1}{2}f\left(\frac{1}{2}\right)+\frac{1}{2}f(1)+\frac{1}{2}f\left(\frac{3}{2}\right)+\frac{1}{2}f(2)$$

$$=\frac{1}{2}f\left(\frac{5}{4}\right)+\frac{1}{2}f(2)+\frac{1}{2}f\left(\frac{13}{4}\right)+\frac{1}{2}f(5)=5.75$$

The option of choosing the right end point of the height of a rectangle is not unique and its left end point can also be considered. Such an alternative gives the graph shown in Fig. 12.4 and estimated area.

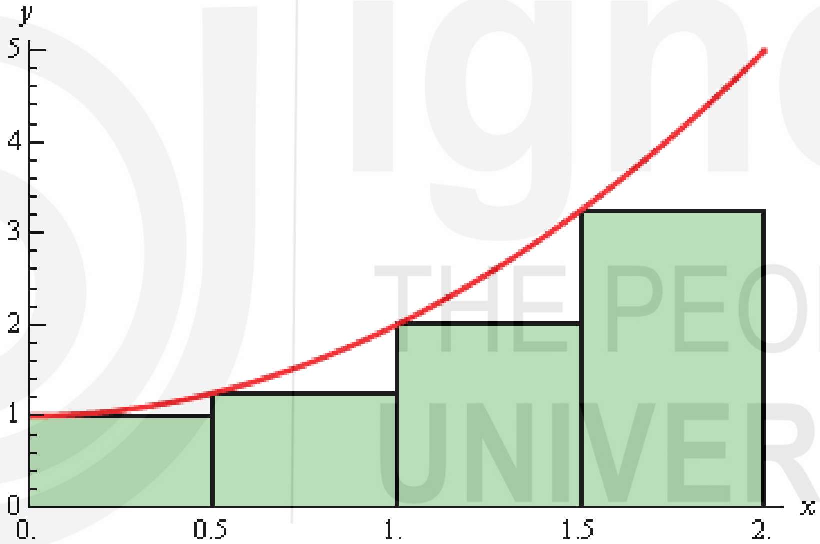

Fig. 12.4: Finding area using left end point of the outer rectangles

$$A_l = \frac{1}{2}f(0)+\frac{1}{2}f\left(\frac{1}{2}\right)+\frac{1}{2}f(1)+\frac{1}{2}f\left(\frac{3}{2}\right)$$

$$=\frac{1}{2}f(1)+\frac{1}{2}f\left(\frac{5}{4}\right)+\frac{1}{2}f(2)+\frac{1}{2}f\left(\frac{13}{4}\right)=3.75$$

In this case, the estimation is an underestimation since each rectangle misses some of the area each time. Instead of using the right or left endpoints of each sub interval, option of taking the midpoint of each subinterval as the height of each rectangle yields the graph shown in Fig. 12.5.

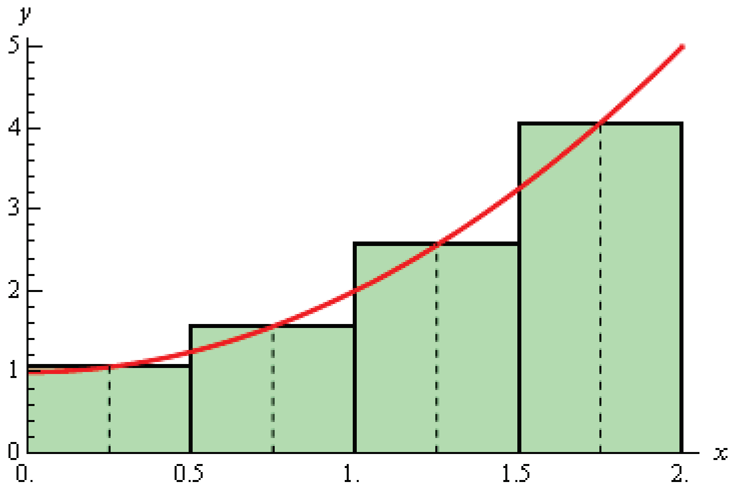

Fig. 12.5: Finding area using mid points of the outer rectangles

The estimation of the area in case of mid-point height of the rectangle is

$$A_1 = \frac{1}{2}f\left(\frac{1}{4}\right)+\frac{1}{2}f\left(\frac{3}{4}\right)+\frac{1}{2}f\left(\frac{5}{4}\right)+\frac{1}{2}f\left(\frac{7}{4}\right)$$

$$=\frac{1}{2}f\left(\frac{17}{16}\right)+\frac{1}{2}f\left(\frac{25}{16}\right)+\frac{1}{2}f\left(\frac{41}{16}\right)+\frac{1}{2}f\left(\frac{65}{16}\right)=4.625$$

Estimating the area by the approximation method produces three estimates viz., right corner, left corner and mid-point height values. Moreover, finding the approximate area through these three options is also influenced by the nature of curve and whether it is increasing or decreasing. For a better result, you need to take more rectangles (i.e., increase n) that will result in minimizing the features of including the curve areas above the curve in the option of right corner height and below the curve areas in the option left corner height. For a more general result, we look into functions that produce curves of various types that may lie above or below that x-axis as well. Doing that leads to our adoption of the method of definite integral to find the area under a curve.

Let us start out with a continuous function $f(x) \geq 0$ on [a, b]. Note that by specifying the interval we are considering a situation that has start and end points. If we divide the interval into n sub intervals each of length we get, $\Delta x = \frac{b-a}{n}$

The end points of each subinterval are,

$$x_0 = a$$

$$x_1 = a + \Delta x$$

$$x_2 = a + 2\Delta x$$

$$x_i = a + i\Delta x$$

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$$x_{n-1} = a + (n-1)\Delta x$$

$$x_n = a + n\Delta x = b$$

Thus, in each interval we have $[x_0, x_1], [x_1, x_2], \ldots, [x_{i-1}, x_i], \ldots, [x_{n-1}, x_n]$. From each we choose a point $x_1^*, x_2^*, \ldots, x_n^*$ such that these will give the height of the rectangles in each subinterval. Seen in terms of a graph, the formulation will look like Fig. 12.6.

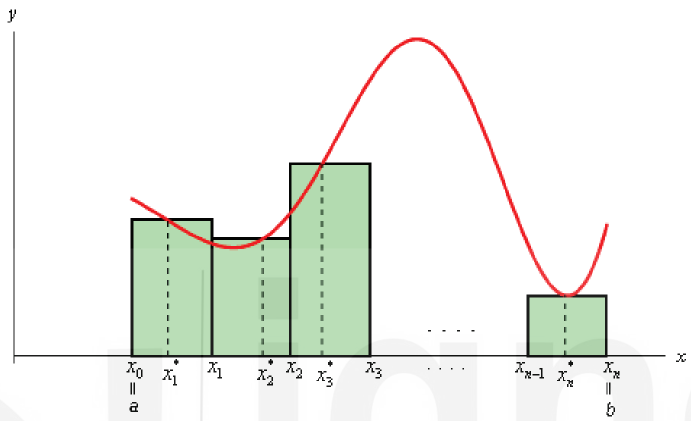

Fig. 12.6: Finding area in a given interval

The area under the curve on the given interval is then approximately,

$$A \approx f(x_1^*)\Delta x + f(x_2^*)\Delta x + \ldots + f(x_i^*)\Delta x + f(x_n^*)\Delta x$$

So $A \approx \sum_{i=1}^{n} f(x_i^*)\Delta x$

The summation above is called a **Riemann** Sum and $A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$ gives the exact area under the curve.

12.8.2 Definition of Definite Integral

Given a continuous function f(x) on the interval [a, b], if we divide the interval into n subintervals of equal width, Δx, and from each interval choose a point, x_i^*, then the **definite integral of f(x) from a to b** is

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

Example 12.22: Using the formula given above, compute of the exact area under $f(x) = x^2 + 1$ on [0, 3].

$$\int_0^3 (x^2+1)dx = \lim_{n\to\infty} \sum_{i=1}^{n} \left[f(x_i) . \frac{(b-a)}{n} \right] = \lim_{n\to\infty} \left(12 + \frac{27}{2n} + \frac{9}{2n^2} \right)$$

$$= \left(12 + \frac{27}{2,\infty} + \frac{9}{2.\infty^2} \right) = 12 + 0 + 0 = 12$$

12.8.3 Properties of Definite Integral

1) $\int_a^b 1dx = b-a$

2) $\int_a^b cf(x)dx = c\int_a^b f(x)dx$ if **c** is a constant.

3) $\int_a^b (f(x)+g(x))dx = \int_a^b f(x)dx + \int_a^b g(x)dx$

4) $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b fx(x)dx$

If the integral of y = 2x + 3 from 1 to 3 is 14 and the integral of the same function from 3 to 5 equals 22, then the integral from 1 to 5 of the function y = 2x + 3 equals 14 + 22, or 36 because the two integrals (from 1 to 3 and 3 to 5) make up the whole integral (from 1 to 5). Conversely, since the integral of $y = 4x^3 - 2x$ from 2 to 4 equal 228, then you know that the sum of the integrals from 2 to 3 and 3 to 4 the same function must also equal 228. If $0 \leq f(x) \leq g(x)$ for all x in [a, b], then $0 \leq \int_a^b f(x)dx \leq \int_a^b g(x)dx$.

5) $\int_a^a f(x)dx = 0$

Example 12.23: The integral of y = 3x + 4 from 2 to 2 because the integral of any function in which both bounds are the exact same number is zero.

6) If a < b then it is convenient to define $\int_b^a f(x)dx = -\int_a^b f(x)dx$

Example 12.24 : If the integral from 1 to 4 of the function $y = x^2$ is 21, then the integral from 4 to 1 of $x^2$is -21 because when the bounds of an integral are switched, the integral is the same, but has the opposite sign (positive changes to negative, and/or negative changes to positive).

Check Your Progress 3

1) What is a definite integral?

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2) What is Riemann Sum?

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3) Why $\int_{a}^{a} f(x)dx = 0$?

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12.9 FUNDAMENTAL THEOREM OF CALCULUS

The processes of differentiation and integration developed quite independently, unrelated to each other. However, the fundamental theorem of calculus established the relationship between differentiation and integration. We have seen from finding the area that the definite integral of a function can be interpreted as the area under the graph of a function. It justifies our procedure of evaluating an anti-derivative at the upper and lower bounds of integration and taking the difference.

The first part of the theorem shows that indefinite integration can be reversed by differentiation. It also proves that for every continuous function, there is an anti-derivative (integral).

Let F be any anti-derivative, or indefinite integral, for **f** on [a, b]. If **f** is a continuous function and is defined by

$$F(x) = \int_{a}^{x} f(t)\,dt$$

Then, $F'(x) = f(x)$

And therefore, $\frac{d}{dx}\int_{a}^{x} f(t)\,dt = f(x)$

Every continuous function f has an anti-derivative, and there is one anti-derivative F(x) given by a definite integral of a f with a variable upper boundary. Varying the lower boundary of the integrand will produce other anti-derivatives. One of the first things to notice about the fundamental theorem of calculus is that the variable of differentiation appears as the upper limit of integration in the integral. This makes sense because if we are taking the derivative of the integrand with respect to x, it needs to be in either (or both) the limits of integration. If we are integrating a function with respect to x, the function we end up with, needs to be in terms of x.

The second part of this theorem tells us how to evaluate a define integral assuming that f has an indefinite integral:

$$\int_{a}^{b} f(x)dx = F(b) - F(a)$$